

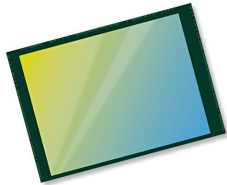
TRULY INNOVATIVE ELECTRONICS

INNOVATION UPDATES

Amongst numerous press releases of new products received by us, these are the ones we found worthy of the title *Truly Innovative Electronics*

Small image sensor for the best low-light performance

Omnivision has launched a 200MP camera sensor, OVB0A, which can be used as primary camera on smartphones. It is one of the smallest sensors featuring 0.56µm pixel size with a 1/3.6cm optical form factor. Also, it



has the best low-light performance in its class. OVB0A features a selective conversion gain for the optimum balance between low-light image quality and high dynamic range. Its on-chip remosaic function enables the sensor to capture 8K videos and 2x crop zoom at 12.5MP. The OVB0A supports CPHY, DPHY, and dual DOVDD (1.8V and 1.2V).

Omnivision

<https://www.ovt.com>

World's smallest bi-directional air velocity sensor

FLS122 from Flusso is the world's smallest air velocity sensor developed exclusively for flow sensing applications. The FLS122 footprint



measures 3.5mm × 3.5mm, making it 80% smaller than any other competitive device available in the market.

The FLS122 is capable of sensing bi-directional flow and provides real-time temperature and air speed measurement of up to 20 metres per second. The sensor can be integrated into an existing product with no flow fixture at low cost. It can be used in a wide variety of space constrained applications, including applications in data centres, gaming PCs that rely on efficient air-cooling to maximise performance and energy-efficiency, and consumer appliances—such as HVACs, air-purifiers, and vacuum cleaners.

Flusso

<https://flussolt.com>

360° lidar sensor to detect small objects

The M1 Edge from Quanergy Systems is the world's first 2D 360° PoE lidar sensor. The sensor offers higher detection accuracy than any other product in



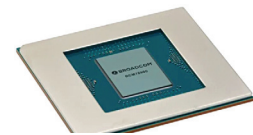
the market. The M1 Edge PoE's angular resolution allows it to detect objects as small as a ball or even a pen, with a high degree of accuracy up to tens of metres away from the sensor. The sensor can enable smart awareness for applications that require simple object detection and alerting capabilities. The M1 Edge PoE is suitable for harsh environments and can be used in a wide variety of applications, including intrusion detection in residential or commercial spaces, perimeter detection, zone protection, and anti-climb prevention.

Quanergy

<https://quanergy.com>

World's first 51.2 Tbit/s Ethernet switch

The BCM78900 is the world's first Ethernet switch capable of 51.2 Tbit/s, or 64 channels of 800Gb Ethernet in data centres from Broadcom. The 5nm



BCM78900 features six on-chip ARM processors, which manage high-bandwidth, fully-programmable streaming telemetry, and provide sophisticated embedded applications like on-chip statistics summarisation. The BCM78900 offers almost double the bandwidth of any other switch silicon available for AI applications. It enables the lowest power and lowest cost for physical connectivity by enabling a direct 100G PAM4 interface to directly attach copper, front panel pluggable optics, and co-packaged optics. It also offers the highest radix of 200GbE ports: 256 ports supported on a single chip, enabling flat, low latency AI/ML clusters.

Broadcom

<https://www.broadcom.com>

The first automotive IMU with ML capabilities

The ASM330LHHX from STMicroelectronics is the first automotive-grade inertial measurement unit (IMU) with ML capabilities. The ASM330LHHX



IMU is a 6-axis module, which houses a 3-axis accelerometer and a 3-axis gyroscope. The IMU module provides